2026

NETWORK512

FORMATIVE ASSESSMENT 3

MOAGI MMOLAWA
20261359

WEEK 1: The IOS Operating System

Topology



User EXEC mode

Terminal

```
Switch> ?
Exec commands:
  connect     Open a terminal connection
  disable     Turn off privileged commands
  disconnect  Disconnect an existing network connection
  enable      Turn on privileged commands
  exit        Exit from the EXEC
  logout      Exit from the EXEC
  ping        Send echo messages
  resume      Resume an active network connection
  show        Show running system information
  ssh         Open a secure shell client connection
  telnet      Open a telnet connection
  terminal    Set terminal line parameters
  traceroute  Trace route to destination
Switch> show ?
  arp                 Arp table
  cdp                 CDP information
  clock               Display the system clock
  crypto              Encryption module
  dtp                 DTP information
  etherchannel        EtherChannel information
  flash:              display information about flash: file system
  history             Display the session command history
  interfaces          Interface status and configuration
  ip                  IP information
  lldp                LLDP information
  mac                 MAC configuration
  mac-address-table   MAC forwarding table
  mls                 Show MultiLayer Switching information
  privilege           Show current privilege level
  sessions            Information about Telnet connections
  ssh                 Status of SSH server connections
  tcp                 Status of TCP connections
  terminal            Display terminal configuration parameters
  users               Display information about terminal lines
  version             System hardware and software status
  vlan                VTP VLAN status
  vtp                 VTP information
Switch> show version
Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen

ROM: Bootstrap program is C2960 boot loader
```

Priviledged EXEC and Global Configuration

```
Switch> enable
Switch# configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)# ?
Configure commands:
  aaa                 Authentication, Authorization and Accounting.
  access-list         Add an access list entry
  banner              Define a login banner
  boot                Boot Commands
  cdp                 Global CDP configuration subcommands
  clock               Configure time-of-day clock
  crypto              Encryption module
  default             Set a command to its defaults
  do-exec             To run exec commands in config mode
  dot1x               IEEE 802.1X Global Configuration Commands
  enable              Modify enable password parameters
  end                 Exit from configure mode
  exit                Exit from configure mode
  hostname            Set system's network name
  interface           Select an interface to configure
  ip                  Global IP configuration subcommands
  line                Configure a terminal line
  lldp                Global LLDP configuration subcommands
  logging             Modify message logging facilities
  mac                 MAC configuration
  mls                 mls global commands
 --More--
```

Configuration Management

```
Switch(config)# end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch# show running-config
Building configuration...

Current configuration : 1080 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
!
interface FastEthernet0/2

Switch# copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Switch# show startup-config
Using 1080 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
```